HREC (MEDICAL) Application form for 2022

UNIVERSITY OF THE WITWATERSRAND, JOHANNESBURG

APPLICATION TO THE HUMAN RESEARCH ETHICS COMMITTEE: (MEDICAL) FOR CLEARANCE OF RESEARCH

Please read the APPENDIX section at the end of this application form before completing this application form. Also please note that no data may be collected before issue of an ethics clearance.

- Category 1 and 2 applications (see below) are subject to monthly closing dates, which may be viewed at https://www.wits.ac.za/research/researcher-support/research-ethics/ethics-committees, chose HREC (Med), see "Downloads"

- Categories 1 and 2 are submitted online at https://www.witsethics.co.za/Login.aspx and one hard copy of the application and all attachments must be submitted to the Ethics Office on the 3rd floor of the PV Tobias Building preferably on the same day as the online application.

- Categories 3,4 and 5 below are not yet online and two completed hard copies of this form may be submitted to the Ethics Office at any time

- Category 7 is not yet online and two completed hard copies of the form referenced below may be submitted to the Ethics Office at any time

Please complete ALL sections of this application form; the Committee needs the information to make a decision. If this is not done, clearance is unlikely - you may have to resubmit a fully completed application.

Categories:

1. New Application:
2. Resubmission – state protocol no.
3. Case Report
4. Sub-study
5. Lab study
6. Waivers – not this form. Please refer to https://www.wits.ac.za/research/researcher-support/research-ethics/ethics-committees/ chose HREC (Med), see "Downloads"

SECTION 1

PRINCIPAL INVESTIGATOR/S

TITLE: Prof **NAME:** Matthew Francis **SURNAME:** Chersich

WITS STAFF NUMBER: A0015316 **WITS STUDENT NUMBER:**

PROFESSIONAL STATUS OR STUDENT YEAR OF STUDY AND DEGREE Professor

UNIVERSITY DEPARTMENT / DIVISION: Wits Reproductive Health and HIV Institute, Faculty of Health Sciences

NON-WITS SITE / INSTITUTION:
Wits RHI, South Africa, Johannesburg
Centre for Sexual Health and HIV AIDS Research Zimbabwe (CeSHHAR), Zimbabwe, Harare
IBM Research Africa, South Africa, Johannesburg
University of Washington, United States, Seattle
University of Cape Town, South Africa, Cape Town
University Peleforo Gon Coulibalu, Côte d'Ivoire, Abidjan
University of Michigan HE2AT United States Ann Arbor

DETAILS OF WHERE STUDY WILL BE DONE

WHERE WILL THE RESEARCH BE CARRIED OUT: The Study covers two large cities in two regions of Africa (Johannesburg, South Africa, Southern Africa and Abidjan, Côte d'Ivoire, West Africa) . This ethics application only applies to Johannesburg. We are not requesting for approval in Abidjan.

HOSPITAL/INSTITUTION WHERE EMPLOYED: Wits Reproductive Health and HIV Institute, Faculty of Health Sciences

FULL-TIME OR PART-TIME EMPLOYEE: Full-time **HPCSA NO:** mp0508365

SECTION 2

CONTACT PERSON DETAILS:

TITLE: Mr **NAME:** Craig

SURNAME: Parker **TEL NO:** +27 79 119 4863

CELL NO: +27 284 8593 **EMAIL:** cparker@wrhi.ac.za

+27 79 284 8593

SUPERVISOR DETAILS:

Title	Name	Surname	Tel No	Cell No	Email

CO-INVESTIGATOR DETAILS:

Title	Full Name	Surname
Dr	Gloria	Maimela
Mr	Craig John	Parker

TRAINING:

ETHICS TRAINING:

Ethics training is a compulsory requirement for consideration of this application. Refer to the DOH2015 guidelines for online ethics training sites at http://nhrec.health.gov.za/index.php/grids-preview, DOH 2015 Ethics, page 56. Please attach copies of your certificates.

Full Name	Ethics Course Name	Date of Ethics course
Matthew Francis Chersich	TRREE Introduction to Research Ethics and Informed Consent	27/05/2022
Craig Parker	TRREE Introduction to Research Ethics	17/05/2022
Gloria Maimela	TRREE Introduction to Research Ethics	09/09/2020

ONLY FOR INVESTIGATOR-INITIATED CLINICAL TRIALS:

GOOD CLINICAL PRACTICE (GCP) TRAINING DETAILS FOR ALL INVESTIGATORS
(Please note that investigators' meetings do not qualify as GCP training and GCP training often includes no ethics component)

Full Name	Ethics Course Name	Date of Ethics course
Matthew Francis Chersich	SA GCP 2020	18/05/2022

Craig Parker	SA GCP 2020	16/05/2022
Gloria Maimela	SA GCP Basic Online Course	09/08/2020

3 TITLE

Research Project 2 (RP2): Innovative machine learning and multi-source data analysis towards development of an urban heat-health Early Warning System for African cities

3.1 PURPOSE OF THE RESEARCH

Postgraduate: Degree / diploma : No
Undergraduate: Degree / diploma : No
Not for Degree Purposes: Yes

State which Degree/Diploma :
NA

3.2 OBJECTIVES OF THE RESEARCH (please list): *(Do not say see attached!)*

The overarching goal of this Research Study is to advance understanding of complex spatially and demographically stratified heat-health interactions in large African cities and to apply this information to develop locally relevant and risk-stratified Early Warning Systems.

More specifically, this overarching goal will be pursued through three aims:

1) Map intra-urban heat vulnerability and exposure across urban areas in large African cities (Aim1)

a) Quantify intra-urban socio-economic and environmental vulnerability (Aim 1a)

b) Develop high resolution urban temperature hazard maps (Aim 1b)

2) Develop a spatially and demographically stratified heat-health outcome forecast model in order to predict the probability of adverse health outcomes at different temperature thresholds (Aim 2)

3) Develop an Early Warning System reflective of geospatial and individualized risk patterns (Aim 3)

a) Develop an EWS based on a digital App (Aim 3a)

b) Develop a “tiered” EWS in collaboration with the Ministries of Health, National Meteorological and Hydrological Services, and other relevant stakeholders (Aim 3b)

3.3 SUMMARY OF THE RESEARCH (give a brief outline of the research plan such that reviewers understand what will be done. (*Do not say see attached!*):

Rapid urban growth, significant levels of informality and increasingly stretched health services, intersecting with observed past and projected future temperature increases, have resulted in a critical intersection between development patterns and climate change in African cities. High ambient temperatures can cause considerable morbidity and mortality in urban areas. The magnitude and pattern of health impacts are determined not just by temperature level, but also by geographical, topographical, socio-environmental and demographic factors. Understanding this complexity is key to developing effective responses that reflect the spatial and demographic heterogeneity of cities.

This Project, to take place in Abidjan, Ivory Coast and Johannesburg, South Africa, is the most detailed exploration of heat and urbanity in Africa to date. We will deploy data science methods ranging from natural language processing to geospatial analysis using predictive analytics to integrate and interrogate multiple diverse data streams alongside conventional health data.

We will apply and further develop existing approaches to image processing, in particular convolution neural networks, to available satellite imagery to construct maps of urban typology including estimates of building types and densities, and green areas. Open mapping data from OpenStreetMaps will also be integrated to categorize street density and layouts. Recently developed high resolution climate model data will be combined with available meteorological station observations, including community observations, temperature variations with altitude, and land surface maps, to develop high resolution vulnerability-hazard maps under different large-scale weather conditions for each city.

Once vulnerability-hazard maps are constructed, health outcomes data will be overlain from a range of sources. Attention networks, which can integrate time lagged relationships, will be used to develop statistical models of a range of heat-health outcomes, beginning with data from large HIV trials in Johannesburg, where residence of participants is known, allowing for fine resolution geospatial mapping. The approach will then be tailored to the available data in Abidjan.

The Project aligns closely with the DS-I Africa objectives, especially its solution focus, possibilities for expansion to other Research Hubs, and potential for progressive expansion to cities across Africa.

3.4 SCIENTIFIC APPROVAL

Has the protocol been approved a scientific committee, *e.g.* Wits Post Graduate Assessors' Committee?
No

3.5 SELECT STUDY DESIGN

Quantitative
StudyDesignOther
Secondary data analysis

3.6 WILL THE STUDY INVOLVE HUMAN MATERIALS

Yes

HumanData

3.7 IS THE STUDY INTENDED TO INFORM IMPROVED CLINICAL OR THERAPEUTIC PRACTICE

No

SECTION 4

4. DOCUMENTATION REQUIREMENTS

4.1.1 Is this project a secondary analysis of data in an established database?

Yes

Note:If "yes", written consent to access a database from the database gatekeeper plus a list of the data to be recorded (see Procedures) must accompany this application.

4.1.2 Is this a sub-study, using data from an aleady-approved primary study
No
Note:If "yes", written consent to access a database from the database gatekeeper plus a list of the data to be recorded (see Procedures) must accompany this application.

Is this documentation attached? No

Please note: If the applicant is gathering new data, interviewing participants, analysing new samples, etc, this does not meet the HREC (Med) definition of a sub-study

4.2 Is this project a retrospective patient record review? No
Note: for a retrospective review the date of the ethics committee meeting at which the application is considered sets the final date for the patient records. If the initial and / or final date is after the meeting date, the study is prospective. Patient records include ALL DATA collected on patients including blood results and radiographs etc.

What is the initial date for the patient records?

What is the final date for the patient records?

Is this documentation attached? NA

Note: the following must accompany this application – written proof of application for permission from the hospital or clinic CEO to do the study, written permission from the clinical entity in which the patients in which patients records are based, how the patients will be selected, what type of records will be examined, a list of the variables to be extracted (see 6.1).

Is this documentation attached? NA

Note: the following must accompany this application

Written proof of application for permission from the hospital or clinic CEO to do the study :
NA
Written permission from the clinical entity in which the patients in which patients records are based :
NA
How the patients will be selected :
NA
What type of records will be examined :
NA
List of the variables to be extracted (see Procedures) :
NA
Have patients consented for the anonymous use of their data?
NA

4.3 Is this project a prospective patient record review? No
Patient records include ALL DATA collected on patients including blood results and radiographs etc.

What is the initial date for the patient records?

What is the final date for the patient records?

Note: the following must accompany this application – written proof of application for permission from the hospital or clinic CEO to do the study, written permission from the clinical entity in which the patients in which patients records are based, how the patients will be selected, what type of records will be examined, a list of the variables to be extracted (see 6.1).

Note: the following must accompany this application

Written proof of application for permission from the hospital or clinic CEO to do the study :
NA
Written permission from the clinical entity in which the patients in which patients records are based :
NA
How the patients will be selected :

What type of records will be examined :

List of the variables to be extracted (see Procedures) :
NA

4.4 If this project involves prospective studies with drugs at a teaching hospital associated with this University, approval must first be obtained from the Hospital's relevant Committee.

Has application been made?(If not, this application cannot be considered) *NA*

4.5 If radiation or isotopes are to be used in prospective studies , written approval must be obtained from the Director, Radiation and Health Physics Unit (james.larkin@wits.ac.za / 011-717 6931).
Note: for patients these are radiation dosages over and above those for standard diagnosis / therapy

Is this attached? NA

4.6 Is a Participant Information Sheet attached?(For written consent)
(see guidance at https://www.wits.ac.za/research/researcher-support/research-ethics/ethics-committees/ chose HREC (Med), see "Downloads")
NA

Informed Consent Form is attached.(For written consent) NA**.**

If informed consent will be verbal or If informed consent is not considered necessary – a written motivation and justification needs to be attached

4.7 If a questionnaire or interview is to be used in the research, it must be attached.

Is it attached? NA

SECTION 5

5. STUDY POPULATION
5.1 If patients/patient records are being studied, state where and how they are selected:

The primary health datasets will be a collation of datasets containing longitudinal data generated in studies that began enrolling participants in 2010 or thereafter (in Johannesburg and in Abidjan). The datasets are drawn from trials among HIV-infected adults, HIV-uninfected adults and adults participating in trials related to COVID-19 prevention or treatment. These datasets will become available throughout the project and will be added based on amendments to original agreements on each of these trials and ethics approval. We will include all participants and associated data in the trial and not use sampling to select the data.

The following databases will be included:

a) databases that contain population groups of interest(High quality health data which can be mapped onto Climate data)

b) the investigators of the original studies have given permission for use of the data

We will not exclude databases with longitudinal data where participant geolocation is not available. We will include both interventional and non-interventional studies.

5.2 Where the participants are not patients,
Will they be invited to volunteer? NA
Will they be selected? NA
State who is invited to volunteer or how the participants are selected:

NA

Are the participants subordinate to the person doing the recruiting? NA
If yes, justify the selection of subordinate participants:

5.3 Will control patients/participants be used? NA
If yes, explain who they are and how they will be recruited

5.4 What is the age range of participants in the study? Need to be able to select >1 group
>=18 years (18 years and older)
If participants are minors (under 18 years), from whom will consent be obtained?

If participants are in the age range 7-17, at least one Assent Document must be attached NA

5.5 Gender: Both

5.6 Number of patients/participants total number based on number of longitudinal studies we include. At this point we cannot specify how many participants as we do not know how many studies will be added to the database. The objective is to create a high quality "big data" Heat Health database. **controls** N/A

5.7 Will the research benefit the patients/participants in any direct way? No
If yes, explain in what way:

5.8 Will participants receive any remuneration? No
If yes, explain what the remuneration is for and how much will be paid

5.9 Will participation, non-participation or withdrawal from the study disadvantage patients/participants in any way? No
If yes, explain in what way:

SECTION 6

6. PROCEDURES (Throughout Section 6, "Procedures" refers to study, not clinical, procedures)

6.1 Mark research procedure(s) that will be used and attached what is required:

Record review: No

Interview form / questionnaire: No

Self-administered questionnaire: No

Focus group: (questions to be used must be attached. Note: there is no confidentiality in a focus group, participants must be told this) No

Examination: No
Nature:

Drug or other substance administration: No
Nature:

Radiographs: No

Isotope administration: No
Nature:

Blood sampling;
(State amount to be collected and the frequency of sampling)

Age Group	Amount	Frequency	Blood Type

Will a biobank be used in the study No
(***See page 7 regarding biobanks)***

Biopsy: No
Explain:

Other procedures: No
Explain:

Use this space to elaborate on procedures marked above:

This will be a re-analysis of existing data using machine learning methods. Once all the data is in the database the study partners will run data science analyses and machine learning algortihms to understand the relationship between heat and health.

6.2 Is/are procedure/(s) routine for:
diagnosis/management? NA
specific to this research? NA
Identify which of the procedures above are routine for diagnosis and management of patients: and identify those procedures specific to the research;

6.3 Who will carry out the procedure(s)? NA

6.4 When will the research project commence after obtaining ethics clearance, and over what approximate time period will the research be done? NA

DATA COLLECTION START DATE:01-Jan-2010 ESTIMATED STUDY END DATE: 03-May-2022

6.5 For studies being done outside the Gauteng Academic Hospitals, please list the number of studies currently being done by the Principal Investigator, the number of patients per study and where they are being done:
NA

6.6 For applications outside the Gauteng Academic Hospitals.
Is the Application Outside of Gauteng? NA

Is the investigator involved in a clinical: Part-Time / Full-Time capacity at the study site? If so, explain in what way : NA

SECTION 7

7 RISKS OF THE STUDY PROCEDURE(S):

7.1 Please consult the risk table at https://www.wits.ac.za/research/researcher-support/research-ethics/ethics-committees/ chose HREC (Med), see "Downloads" and indicate the level of risk to:

Patients/Participants: Minimum

Research team members: None

All other persons: None

7.2 Please indicate whether the patients/participants will be exposed to any levels of:

Physical discomfort: No
Pain: No
Possible complications: No
Side effects from agents used: No
Breach of confidentiality: Yes
Possible stigmatisation: No
Psychological stress: No
If you have checked any of the above except "No risk" please provide details:

Risk to participate in this study is minimal as there are no new procdures involved and many steps will be taken to protect personal information. Data from partipipants will be safeguarded and only made available to study investigators through strict multi-tiered access control. Personal identifying information such as geo-location wil further be de-identified, encryted and jittered in order to preserve confidentiality. Moreover , no personal details of any participants will be made publicly accessible through this project or published in a journal article.

SECTION 8

8. GENERAL

8.1 For any study, has permission from relevant authority/ies been obtained to do the study? No

State name of authority/ies and provide written proof of application for permission(s):
NA

8.2 Has this study been submitted to other Ethics Committees? Yes
If yes, what is the status of the application? Under Review at UCT Faculty of Science Ethics Review Board

8.3 How will confidentiality be maintained so that participants are not identifiable to persons not involved in the research? Please answer the questions below:

Will data be anonymous? Data may not be fully unidentifiable as we are collecting date of birth and location data, where available. However, we will take steps to minimize the risk to privacy of participants. We will not be collecting names of participants and no identifiable data will be published. In addition, the data will be safeguarded in a password protected server with limited access to study investigators. In addition, we will use principles of minimization, to ensure that only essential data that is required for the study objectives are stored. Lastly, wherever possible, we will anonymize data through three methods:
1. Street addresses will be aggregated into geographical regions such that many records map to the same region and it is no longer possible to derive individual residential locations. Aggregation will also consider the number of records that map to the same geographical area.
2. Latitude/longitude coordinates will be “jittered” which involves adding random values to each coordinate such that

the exact location is lost, while retaining sufficient geographical information to support the analysis. Jittering will be used to shift latitude/longitude locations in the order of 10s of kilometres, adequate to prevent locating individual residential locations.
3. Where Date of Delivery/Date of Birth is considered personally identifiable information, this will be used to associate the relevant climate variables and then will be removed from the dataset.
How?

Will identifiable data be coded and the 'links' kept separate? Data will be de-identified as described above. **How?**

Who will have access to the raw data? Primary encrypted data will be stored in an isolated virtual storage server that is only accessible to private data team. This team consists of Chris Jack from UCT, Sibusisiwe Makhanya from IBM who are responsible for data management core of the project.

In order to control access Climate Systems Analysis Group compute infrastructure sits within the University of Cape Town(UCT) intranet and falls under the broader UCT security policies and services. UCT Information and Communication Technology Services(ICTS) implements a Cisco firewall and Cisco Virtual Private Network service to ensure that intranet access is limited to authorized users.

8.4 To whom will results be made available? (*e.g.* participants, supervisor, hospital management,etc)

DSI-Africa consortium, policymakers, the scientific community , the general public, local, provincial, and national authorities and other stakeholders.

8.5 How will the results be disseminated?(*e.g.* journal article, conference presentation, departmental seminar, etc)

The HE²AT Center will disseminate study findings and other learnings in a timely manner across our consortium, partners, policymakers, the scientific community, other stakeholders, and the general public to maximise its impact. Our engagement plan includes:

o Scientific publications and presentations to scientific conferences
o Community engagement; and
o Policy engagement

Scientific publications from the Research Studies will be published in open access journals with data sharing. The final protocol will be made available on open access to allow replication of the approach in other settings and will be registered with the appropriate registry. It is anticipated that the study will generate high quality data sufficient for at least four publications in high-impact journals. A publication plan will be developed within the consortium that will define the potential publications, authorship, and timeline for manuscript production.

Each year there will be presentations to scientific conferences, such as the International Society for Environmental Epidemiology, the NIH biannual meetings, the EDCTP Forum, and others. Presentations will also be made upon request to meetings organized by the African Regional Office of the World Health Organization, the United Nations Environment Programme, the United Nations Development Programme, and others. Further, study findings and communication materials may be shared via the Global Heat Health Information Network. The emphasis will be on promoting junior researchers to increase their confidence and skill in preparing and delivering scientific presentations. Costs for these presentations are included in the Research Studies.

The HE²AT Center will promote the Studies and their findings among the communities in which the research will be conducted, to ensure increased acceptability of the Studies and their objectives, as well as promote awareness of heat-health risks. Dissemination tools such as newsletters, brochures, and introductory study posters, may be distributed among the community. We will emphasize the guiding principles of Good Participatory Practice, including respect, mutual understanding, integrity, transparency, accountability, and autonomy. Some of the team members already engage frequently with the media about climate or extreme heat concerns, which opens up further avenues for the dissemination of findings.

Study results will be disseminated to the local, provincial, and national authorities, to inform their Climate Change and Health Adaptation Planning. Very often these authorities do not receive sufficient technical support for the design and implementation of the types of solution that will be developed.

The Africa CDC will also play a major role in dissemination activities of the HE²AT Center and are fully committed to participate actively in the activities of the Center, including hosting joint dissemination meetings and/or stakeholder consultations. Timely dissemination and clear communication of results is key for updating guidance and for informing sub-national, national, and international policies. The results will contribute to the development and implementation of heat-health early warning and integrated surveillance systems and associated policies, strategies, and interventions to reduce the health risks of extreme heat in Cote d'ivoire and South Africa, with the potential of broader application for low- and middle-income countries.

8.6 Will there be financial costs to:

Participants: No
Hospital/Institution: No
Other: No

Explain any box marked "Yes":

8.7 If no protocol is attached please indicate: Budget? How will the research be funded?

Please give details of the source of funds

Funding acknowledgement: This grant is supported by the NIH Common Fund, which is managed by the Fogarty International Center (FIC). NIH award number U54TW012083. Protocol is attached.

8.8 Any other information, which may help the Committee to evaluate this application, may be provided here: NA

APPENDIX

1. Please note that it is the responsibility of the Principal Investigator in an application to ensure that he/she has received the final HREC (Medical) Clearance before commending any research. This is signified by, and only by, the issuing of a Clearance Certificate, which will be headed as such.
2. Please indicate clearly, where correspondence should be sent; failure to do this may cause delays. Please provide the supervisor's email address (where applicable) for sending copies of correspondence.
3. It is required that a protocol detailing the background to the research, the design of the investigation and all procedures, is submitted with the application.
4. If any doubt exists, please come into the Ethics Office (Phillip Tobias Building, 3rd Floor, Cnr York Road and Princess of Wales Terrace, Parktown) during normal office hours and ask the staff on duty there
5. Please note that written clearances will not normally be available until 14-28 working days after a Committee meeting .
6. **STORAGE OF BLOOD AND/OR TISSUE SAMPLES:**
   The policy of the ethics committee is:
   - If, blood or tissue specimens are to be stored for future analysis and / or it is planned that analysis may be done outside Wits, then the specimens must be stored at Wits with release of sub-samples only once projects have been approved by the local Research Ethics Committee applicable to where the research will be done, as well as by the Wits Human Research Ethics Committee: (Medical);
   - A separate information sheet and consent form for this is required. Please see the Standard Operating procedure at https://www.witshealth.co.za/Services/Research-Ethics/Operating-Procedures
   - For information on Biobanks and the Biobank Ethics Committee within the Wits Human Research Ethics Committee, please go to https://www.wits.ac.za/research/researcher-support/research-ethics/ethics-committees/ chose HREC (Med), see "Biobanks Ethics Commitee"
   - Only approved analyses may be done;
   - Specimens may not be shared with anyone, unless approved by the Wits Human Research Ethics Committee (Medical).
7. Evaluation of applications from private sites/institutions without any affiliation to Wits may be done but is at the discretion of the Wits Human Research Ethics Committee (Medical). In such instances a processing fee is payable - current cost and payment procedure available on enquiry.
8. Researchers from abroad should obtain ethics clearance BEFORE arriving at Wits, as a tight time schedule is not considered a valid reason for departing from Wits Standard Operating Procedure. A Wits collaborator may help obtain the clearance.
9. Researchers with syndicates in the Wits Health Consortium – please read the home page at www.witshealth.co.za regarding the requirement that the syndicate must be based in a Wits academic department or recognised research entity
10. Please note: No late applications will be accepted after the submission date listed at ; "https://www.wits.ac.za/media/wits-university/research/documents/002.Ethics%20Meeting%20and%20Submission%20Dates.pdf" Applications received after this date will be carried forward to the following meeting; incomplete applications will not be accepted
11. No data may be collected before a clearance is issued by the HREC (Medical).

PLAGIARISM POLICY

The University's policy on plagiarism is set out at:
https://www.wits.ac.za/media/wits-university/faculties-and-schools/-engineering-and-the-built-environment/mechanical-industrial-aeronautical-engineering/documents/postgraduate-documents/Wits%20plagiarismpolicy.pdf
Applicants seeking ethics clearance are required to be familiar with this policy

Signatures: Matthew Francis Chersich

Date: 30 May 2022 Applicant's Signature: 

WHO WILL SUPERVISE THE PROJECT? (Where applicable)

Name Department:

Telephone No: Email:

Signature: Date:

HEAD / RESEARCH COORDINATOR OF DEPARTMENT / ENTITY IN WHICH STUDY WILL BE CONDUCTED (Where applicable) (Wits Students Academic HOD must sign)

Name: Prof Helen Rees Department / Entity Wits RHI
Tel No: 011 358 5500 Email: hrees@wrhi.ac.za

Signature:  Date: 3 June 2022